

Semester Test 1

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Electricity and Electronics EBN 122

23 August 2019

Test Information

Maximum marks:	46	Full marks:	45
Duration of test:	90 min + (5 min before and 10 min after for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)		18	

Important

1. The examination regulations of the University of Pretoria apply.
2. Answer all questions on the OCR forms provided. The question number **01** in this question paper refers to row 01 of the OCR form.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

Lecturers: Prof J.P. Jacobs, Prof J.P. de Villiers, Dr X. Ye

INSTRUCTIONS**Do step 0 in the first 5 minutes of the session. (-5 to 0 min)****Do step 1 in the next 90 minutes of the test. (0-90 min)****Do step 2 within the next 10 minutes. (90-100 min)**

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2019EBN122S01**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1

Questions 1 to 3 [3]

A lightning bolt induces a constant current of 10 kA for 15 μ s as it strikes the ground.	
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- | | |
|----|--|
| 01 | How much charge Q is transferred by the lightning bolt? [1] |
| 02 | If the voltage of the bolt is 10^9 V with reference to the ground, how much power P is dissipated by the lightning bolt during the strike? [1] |
| 03 | Calculate the total energy E of the lightning bolt in Wh. [1] |

01 $Q = 10 \times 10^3 \times 15 \times 10^{-6} = 150 \times 10^{-3}$
 $= 150 \text{ mC} \rightarrow$

02 $P = VI = 10^9 \times 10 \times 10^3$
 $= 10 \times 10^{12} \text{ W} = 10 \text{ TW} \rightarrow$

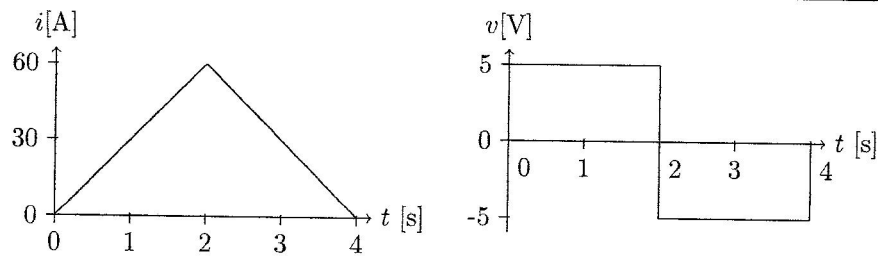
03 $E = \int_0^{15 \times 10^{-6}} 10 \times 10^{12} dt$
 $= 10 \times 10^{12} \times 15 \times 10^{-6}$
 $= 150 \times 10^6 \text{ J} = 150 \text{ MJ}$

Convert to Wh

$E = \frac{150 \times 10^6}{3600} = 41.66 \text{ kWh} \rightarrow$

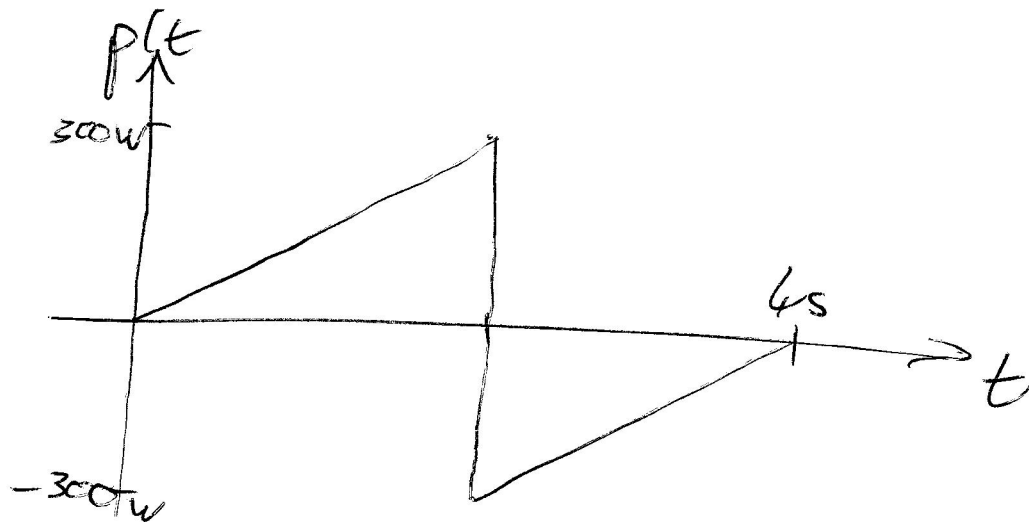
Question 4 [2]

The figure below shows the current through and the voltage across a circuit element.



04 Find the total energy E_e absorbed by the element for the period of $0 < t < 4$ s. [2]

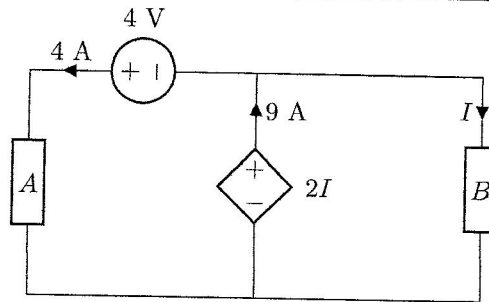
04 $p = vi$



$$E_e = \int_0^4 p(t) dt = 300 - 300 = 0 \text{ J} \rightarrow$$

Questions 5 to 7 [3]

Calculate the power relating to the following elements in the circuit below:



05	Calculate the power P_D associated with the dependent voltage source. [1]
06	Is the dependent voltage source in the circuit supplying power (write a 1) or absorbing power (write a 2)? [1]
07	Calculate the resistance R_B of circuit element B. [1]

05. $P_D = -V_i = -(2 \times 5)(9) = -90 \text{ W}$ $\rightarrow$

06. 1 - Supplying power

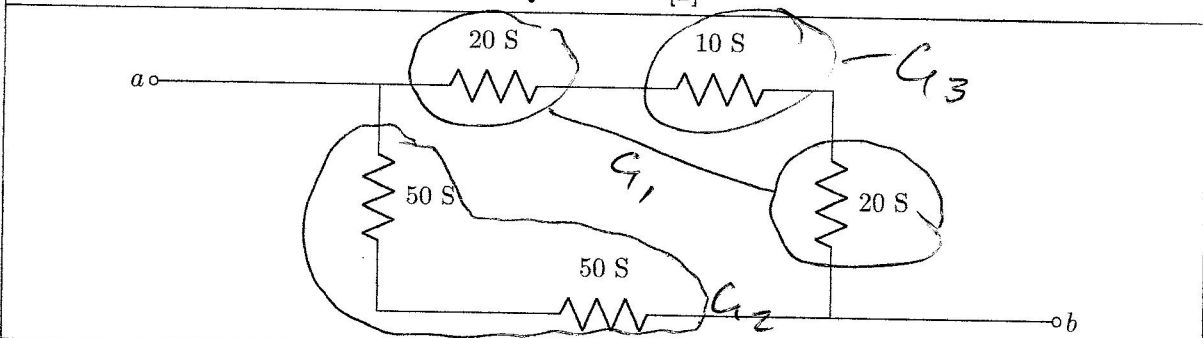
07. $V = IR \quad \therefore R_B = \frac{V}{I} = \frac{10}{5}$

$(I = 5 \text{ A because of KCL,}$
 $\text{i.e. } 9 = 4 + I)$ $= 2 \Omega$

Total Section 1: [8]

Section 2

Question 8 [2]



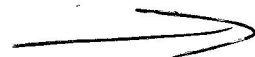
08 Determine the equivalent conductance G_{ab} between nodes a and b . [2]

$$G_1 = \frac{20 \times 20}{20 + 20} = 10 \text{ S}$$

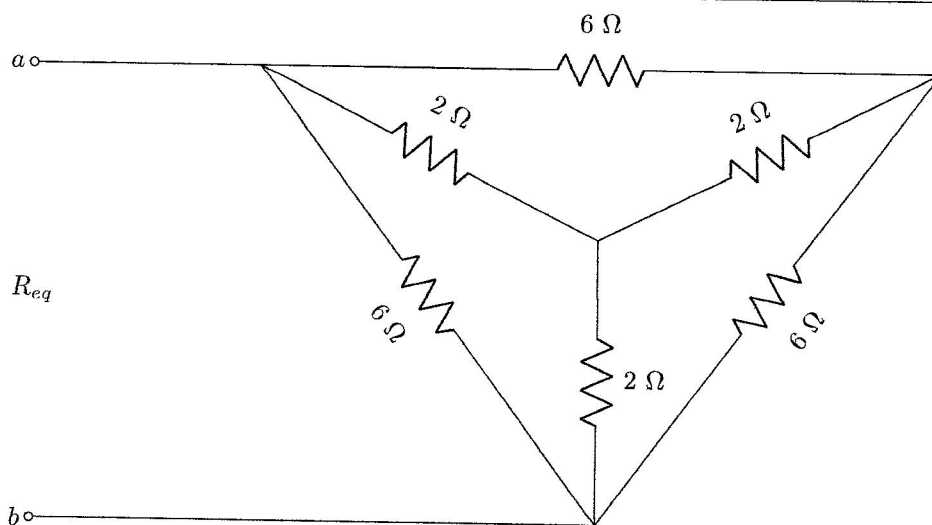
$$G_2 = \frac{50 \times 50}{50 + 50} = 25 \text{ S}$$

$$G_1 \text{ in series with } G_3 \therefore \frac{10 \times 10}{10 + 10} = 5 \text{ S}$$

$$G_2 // 10 = G_2 + 10 = 25 + 5 = 30 \text{ S}$$

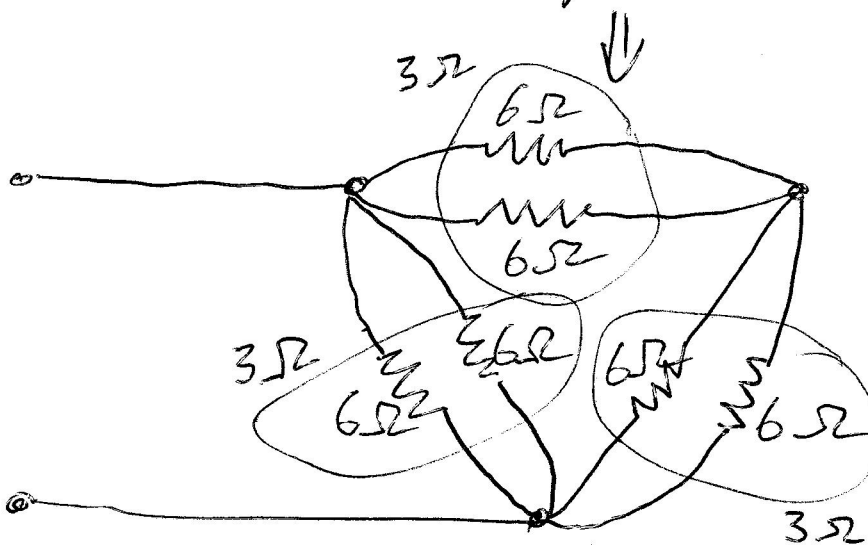


Question 9 [2]



09 Determine the equivalent resistance R_{eq} between nodes a and b . [2]

Transform inner Wye to Delta



$$6//6 = \frac{6 \times 6}{6 + 6} = \frac{36}{12} = 3\Omega$$

$$\begin{aligned}
 R_{eq} &= 6//6 // (6//6 + 6//6) \\
 &= 3 // (3 + 3) = 3//6 \\
 &= \frac{3 \times 6}{3 + 6} = 2\Omega
 \end{aligned}$$

Questions 10 to 11 [4]	
Consider the circuit below and assume that $V_1/V_2 = 3$ and $R_2 = 20 \Omega$.	
10	Calculate R_1 . [2]
11	Calculate I_a . [2]

10 Voltage dividers: $V_1 = \frac{R_1}{R_1 + R_2} (40)$

$$V_2 = \frac{R_2}{R_1 + R_2} (40)$$

$$\frac{V_1}{V_2} = 3 = \frac{R_1 \cancel{(40)}}{\cancel{(R_1 + R_2)} R_2 \cancel{(40)}} = \frac{R_1}{R_2} = 3$$

$$R_1 = 3R_2 = 3(20) = 60 \Omega \Rightarrow$$

11 $\frac{V_1}{V_2} = 3$ and $V_1 + V_2 = 40$ (2)

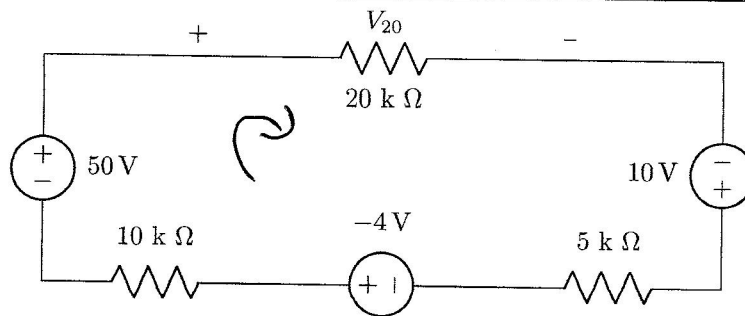
$$V_1 = 3V_2 \quad (1)$$

$$(1) \text{ in } (2): 3V_2 + V_2 = 40 \therefore V_2 = 10V$$

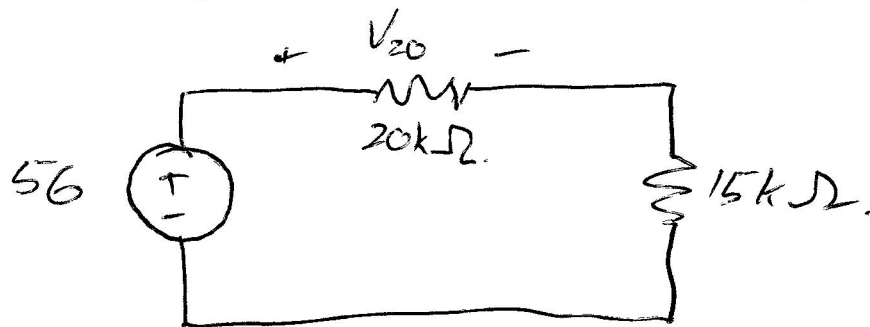
$$I_s = 0.25V_2 = 2.5A$$

$$\text{Current division } I_a = \frac{20}{20+30} I_s = 1A \Rightarrow$$

Question 12 [2]

12 Determine V_{20} . [2]

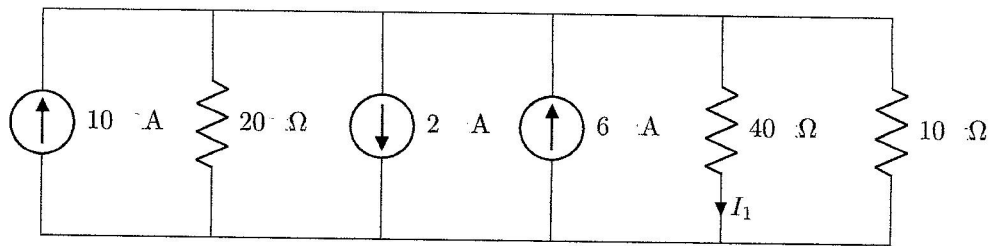
Total voltage sources : $50 + 10 - 4 = 56V$.



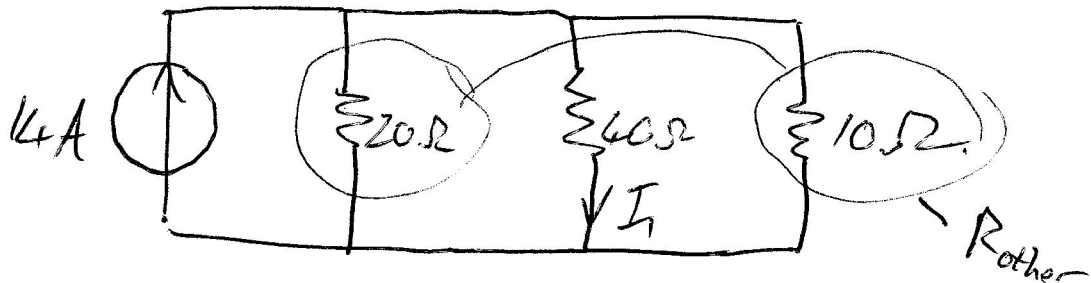
Voltage divider :
$$V_{20} = \frac{20k}{20k + 15k} (56)$$

$$= 32V$$


Question 13 [2]

13 Determine I_1 . [2]

All elements are in parallel $\rightarrow$ replace all current sources with single source



Current division:

$$R_{other} = \frac{20 \times 10}{20 + 10} = 6.667 \Omega$$

$$I_1 = \frac{R_{other}}{(R_{other} + 40)} 14$$

$$= \frac{6.667}{(6.667 + 40)} 14 = 2 \text{ A} \rightarrow$$

Total Section 2: [12]

Section 3

Question 14 [2]

14 Determine θ if $v_1 = -2$. [2]

$$\begin{bmatrix} -2\theta & -4 \\ 6 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -2\theta \\ 12 \end{bmatrix}$$

14 Eq 1 $-2\theta v_1 - 4v_2 = -2\theta$
 $-2\theta(-2) - 4v_2 = -2\theta$
 $4\theta - 4v_2 = -2\theta$
 $-4v_2 = -6\theta$
 $v_2 = \frac{3}{2}\theta \quad (1)$

Eq 2 $6v_1 + 4v_2 = 12$
 $6(-2) + 4\left(\frac{3}{2}\right)\theta = 12$
 $-12 + 6\theta = 12$
 $6\theta = 24$
 $\theta = 4 \rightarrow$

$$(v_2 = 6V)$$

Question 15 [2]

15 Determine σ if $i_3 = -1.2$. [2]

$$\begin{bmatrix} 2 & 1 & 4 \\ 0 & -2 & 1 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} \sigma \\ 4 \\ -2 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 2 & 1 & 4 \\ 0 & -2 & 1 \\ 2 & 4 & 0 \end{vmatrix} = (2) + (0) + (2) - (-16) - (8) - 0 = -10$$

$$\Delta_3 = \begin{vmatrix} 2 & 1 & 0 \\ 0 & -2 & 4 \\ 2 & 4 & -2 \end{vmatrix} = (8) + 0 + (8) - (-40) - (32) - (0) = 40 - 16 = 16$$

$$i_3 = \frac{\Delta_3}{\Delta} = \frac{40 - 16}{10} = -1.2$$

$$\therefore 40 - 16 = -12$$

$$40 = 4$$

$$\sigma = 1$$



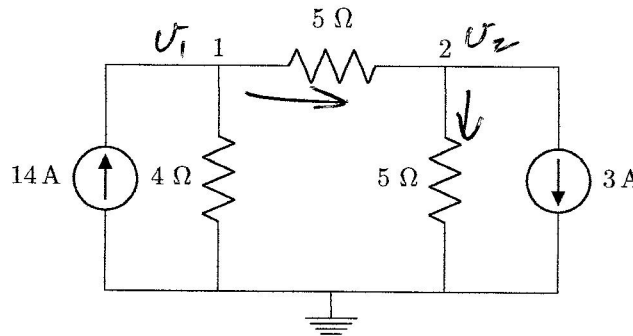
Questions 16 to 19 [4]

Use nodal analysis to set up the following two equations of the form:

$$av_1 + bv_2 = 280$$

$$cv_1 + dv_2 = 15$$

Calculate the unknown coefficients a , b , c and d . Note: Coefficients should be integer numbers.

16 a [1]17 b [1]18 c [1]19 d [1]

Node 1

$$\frac{v_1}{4} + \frac{v_1 - v_2}{5} = 14$$

$$5v_1 + 4v_1 - 4v_2 = 280$$

$$9v_1 - 4v_2 = 280$$

Node 2

$$\frac{v_1 - v_2}{5} = \frac{v_2}{5} + 3$$

$$v_1 - v_2 = v_2 + 15$$

$$v_1 - 2v_2 = 15$$

$$a = 9 \quad b = -4 \quad c = 1 \quad d = -2$$

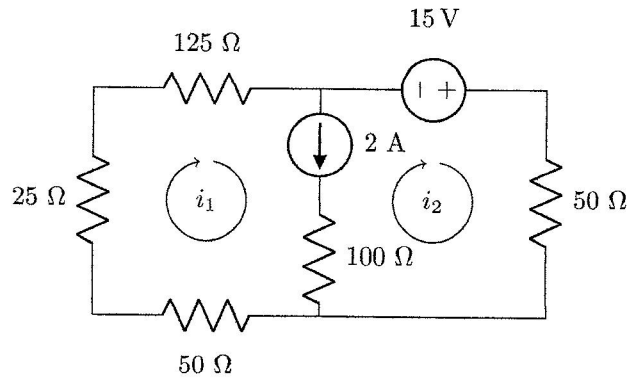
Questions 20 to 23 [4]

Use mesh analysis to set up the following two equations of the form:

$$ei_1 + fi_2 = 15$$

$$gi_1 + hi_2 = 2$$

Calculate the unknown coefficients e , f , g and h . Note: Coefficients should be integer numbers.



20	e [1]	21	f [1]	22	g [1]	23	h [1]
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Mesh 1+2
(supermesh)

$$25i_1 + 125i_1 - 15 + 50i_2 + 50i_1 = 0$$

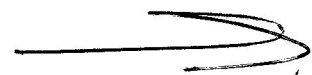
$$200i_1 + 50i_2 = 15 \quad (1)$$

Middle branch

$$i_1 - i_2 = 2 \quad (2)$$

$$\therefore e = 200 \quad f = 50$$

$$g = 1 \quad h = -1$$



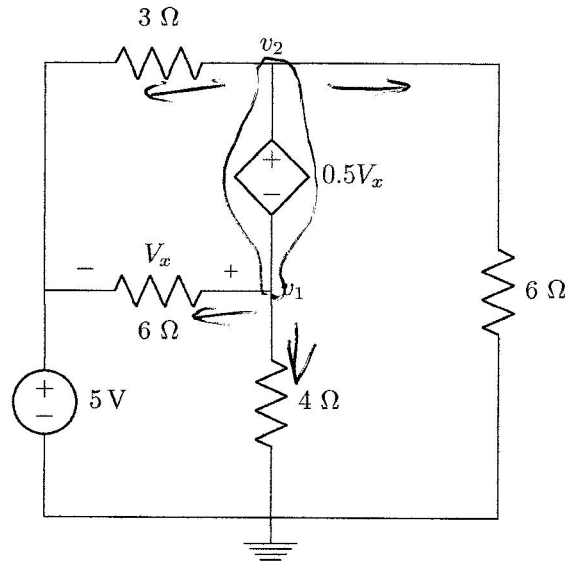
Questions 24 to 27 [4]

Use nodal analysis to set up the following two equations of the form:

$$qv_1 + rv_2 = 30$$

$$13v_1 + sv_2 = t$$

Calculate the unknown coefficients q , r , s and t . Note: Coefficients should be integer numbers.

24 q [1]25 r [1]26 s [1]27 t [1]

Supernode KCL: $\frac{v_2 - 5}{3} + \frac{v_2}{6} + \frac{v_1 - 5}{6} + \frac{v_1}{4} = 0$

$$2v_2 - 10 + v_2 + v_1 - 5 + 1.5v_1 = 0$$

$$2.5v_1 + 3v_2 = +15$$

$$5v_1 + 6v_2 = 30$$

Relationship between v_1 and v_2 :

$$v_2 = v_1 + 0.5v_{x_c}$$

$$q = 5$$

$$v_2 = v_1 + 0.5(v_1 - 5)$$

$$r = 6$$

$$-1.5v_1 + v_2 = -2.5$$

$$s = -8.666$$

$$1.5v_1 - v_2 = 2.5$$

$$t = 21.666$$

$$(x. 8.666)$$

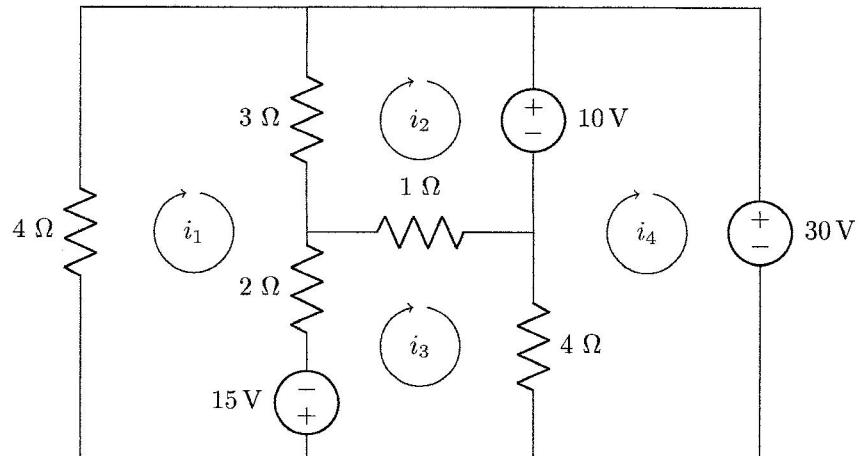
$$13v_1 - 8.666v_2 = 21.666$$

Questions 28 to 31 [4]

Use inspection to compile a matrix equation of the form below

$$\begin{bmatrix} 9 & -3 & -2 & 0 \\ -3 & v & -1 & 0 \\ -2 & -1 & 7 & w \\ 0 & 0 & w & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 15 \\ x \\ -15 \\ y \end{bmatrix}$$

and provide the values of the matrix elements v , w , x and y .

28 v [1]29 w [1]30 x [1]31 y [1]

$$v = 4$$

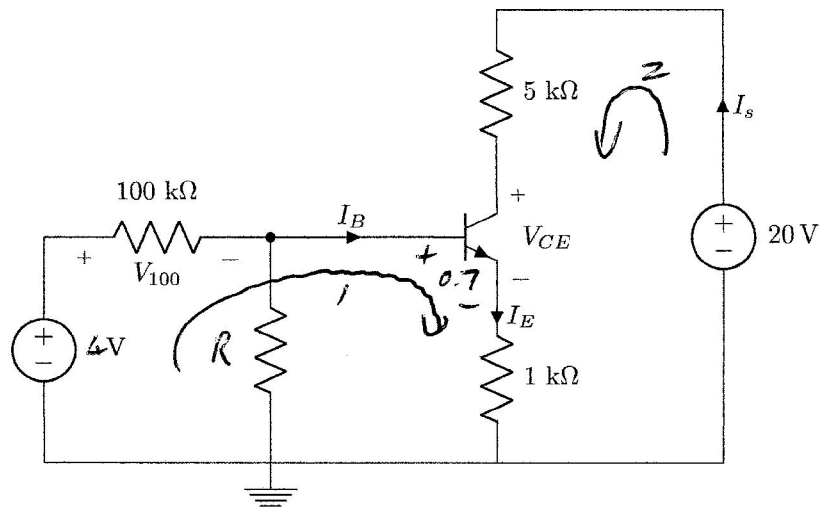
$$w = -4$$

$$x = -10$$

$$y = -20$$

Questions 32 to 35 [6]

Consider the case where $I_E = 1.01 \text{ mA}$ and $\beta = 100$ in the circuit below. Let $V_{BE} = 0.7 \text{ V}$, i.e. the transistor is operating in the active region. Determine I_B , V_{100} , I_s and V_{CE} .

32 I_B [1]33 V_{100} [2]34 I_s [1]35 V_{CE} [2]

$$32 \quad I_E = (\beta + 1) I_B$$

$$1.01 \times 10^{-3} = 101 I_B$$

$$I_B = 10 \times 10^{-6} \text{ A} = 10 \mu\text{A}$$

$$33 \quad \text{KVL ground loop 1}$$

$$-4 + V_{100} + 0.7 + I_E (1000) = 0$$

$$V_{100} = 3.3 + 1.01 \times 10^{-3} \times 1 \times 10^3 = 0$$

$$V_{100} - 3.3 + 1.01 = 0; V_{100} = 2.29 \text{ V}$$

$$35 \quad I_s = I_C = \beta I_B = 1 \times 10^{-3} \text{ A} = 1 \text{ mA}$$

$$36 \quad \text{KVL loop 2}; -20 + I_s \times 5000 + 1000 I_E + V_{CE} = 0$$

$$V_{CE} + 5 + 1.01 = 20 \quad V_{CE} = 13.99 \text{ V}$$